Q proves numeral sums equal the numeral of the sum

The Open Logic Project contributors

openlogic:incompleteness/representability-in-q:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-incompleteness-representability-in-q/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: complete two-case derivation; nothing delegated

Editorial note. The label lem:q-proves-add names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Q axioms Q_4 and Q_5. Q_4: x + 0 = x; Q_5: x + y' = (x + y)'.. Base case is Q_4; the inductive step is Q_5 plus the induction hypothesis.

Lemma 0.1. $\mathrm{Th}Q \vdash (\overline{n} + \overline{m}) = \overline{n + m}$

Proof. We prove this by induction on m. If $m = 0$, the claim is that $\mathrm{Th}Q \vdash (\overline{n} + 0) = \overline{n}$. This follows by axiom Q_4. Now suppose the claim for m; let's prove the claim for $m + 1$, i.e., prove that $\mathrm{Th}Q \vdash (\overline{n} + \overline{m + 1}) = \overline{n + m + 1}$. Note that $\overline{m + 1}$ is just $\overline{m}'$, and $\overline{n + m + 1}$ is just $\overline{n + m}'$. By axiom Q_5, $\mathrm{Th}Q \vdash (\overline{n} + \overline{m}') = (\overline{n} + \overline{m})'$. By induction hypothesis, $\mathrm{Th}Q \vdash (\overline{n} + \overline{m}) = \overline{n + m}$. So $\mathrm{Th}Q \vdash (\overline{n} + \overline{m}') = \overline{n + m}'$. □